

Scalability & Future Plan

Date	6 July 2026
Team ID	XXXXXX
Project Name	Smart Lender – Loan Approval Prediction System
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the **Smart Lender – Loan Approval Prediction System** can be enhanced to support a larger number of users, improve system performance, strengthen security, and introduce new features in future versions. It also presents the team's roadmap for expanding the application's capabilities beyond the current implementation.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	No user authentication system	Anyone can access the application without login.	High
2	No database to store prediction history	Previous loan predictions cannot be saved or retrieved.	Medium
3	Uses a basic Machine Learning model	Prediction accuracy can be improved with advanced models.	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports limited concurrent users	Deploy on a scalable cloud platform with load balancing.
2	Data Storage	No database integration	Integrate MySQL or PostgreSQL to store user and prediction data.
3	Performance	Single Flask application	Optimize model loading, caching, and backend processing for faster predictions.
4	Security	Basic input validation	Add user authentication, HTTPS, and secure data handling.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Add user authentication and account management	2–3 Months	Secure access and personalized user experience.
Phase 3	Store prediction history using a database	3–4 Months	Enable users to view and manage previous loan predictions.
Phase 4	Integrate advanced Machine Learning models and personalized loan recommendations	5–6 Months	Improve prediction accuracy and provide intelligent loan suggestions.